



BUSINESS INTELLIGENCE PROJECT

BIKE SHARING

ANALYTICS

SQL

Power BI

DAX

Power Query

Data Modeling



What, Why & Business Objective



WHAT

An end-to-end Business Intelligence project analyzing Capital Bikeshare data using SQL, Power BI, and DAX to uncover actionable insights.



WHY

Bike-sharing operators need data-driven decisions to optimize fleet, pricing, and marketing — especially around demand fluctuations and weather.



GOAL

Deliver an interactive Power BI dashboard that surfaces KPIs, revenue trends, seasonal patterns, and customer segmentation at a glance.

730

Days of Data

3M+

Rider Records

20+

SQL Queries

10

Dashboard KPIs

What Problems Need Solving?



Rider Demand Analysis

When do peak demand periods occur?
What triggers high vs low ridership days?



Revenue Fluctuations

Revenue is inconsistent across months.
What seasonal patterns drive revenue swings?



Weather & Seasonal Impact

How much does weather condition affect ridership, and which seasons underperform?



Customer Segmentation

What's the ratio of casual vs registered users? Are loyalty trends improving?



Operational Decisions

How can management optimize operations based on daily and weekly demand patterns?



Profitability Gaps

Which months, quarters, and weather conditions create the biggest profit gaps?

Capital Bikeshare — Data Summary

730

Total Days

3M+

Rider Trips

16

Feature Columns

2 Yrs

Time Span

Column	Description	Type	
date	Trip date	DATE	Temporal Date, Month, Quarter, Season, Day Name
casual_users	Unregistered riders	INT	Rider Types Casual, Registered, Total, Loyalty Ratio
registered_users	Member riders	INT	
total_users	All daily riders	INT	Financial Revenue, Operating Cost, Profit
weather_situation	1=Clear to 4=Extreme	INT	Weather Weather Situation → Clear, Cloudy, Rainy, Extreme
season / month	Temporal grouping	VARCHAR	
revenue / profit	Derived financials	DECIMAL	Operational Demand Level, Rider Type, Day Category

From Raw Data to BI-Ready Tables

```
UPDATE cleaned_capital_bike_share
SET demand_level =
CASE
  WHEN total_users < 1000 THEN 'Low Demand'
  WHEN total_users BETWEEN 1000 AND 5000 THEN 'Medium Demand'
  ELSE 'High Demand'
END;
```

```
SELECT
  ROUND(SUM(revenue),2) AS total_revenue,
  ROUND(SUM(profit),2) AS total_profit
FROM cleaned_capital_bike_share;
```

```
SELECT
  date,
  total_users,

  ROUND(
    AVG(total_users)
    OVER(
      ORDER BY date
      ROWS BETWEEN 6 PRECEDING
      AND CURRENT ROW
    ),2
  ) AS rolling_avg_7_days

FROM cleaned_capital_bike_share;
```



Schema Enrichment

Added 10+ derived columns: revenue, profit, demand_level, weather_label, rider_type, loyalty_ratio, season_name



KPI Calculations

Revenue = (casual × ₹100) + (registered × ₹60). Profit = Revenue – Fixed Cost (₹20,000/day)



Window Functions

RANK(), DENSE_RANK(), LAG/LEAD, 7-day rolling average, cumulative revenue & profit

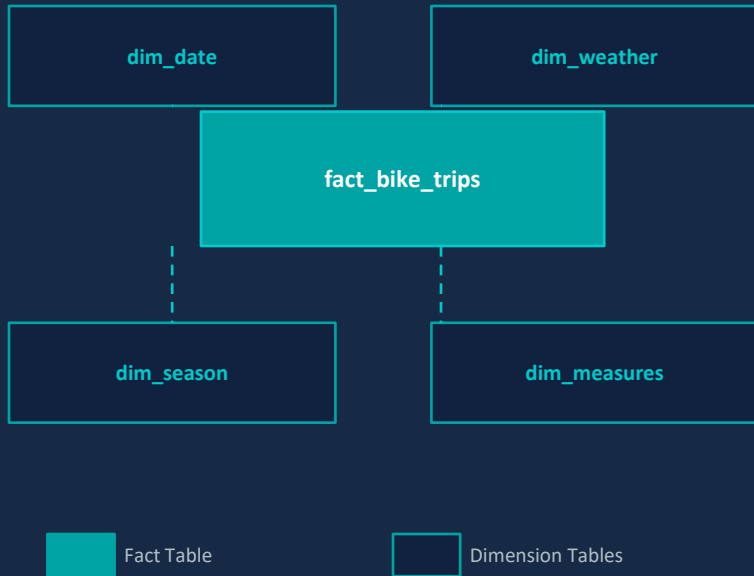


Views Created

10 SQL Views: monthly_summary, weather_analysis, rider_type, demand_level, quarterly_performance, and more

Power BI Model & Calculated Measures

DATA MODEL STRUCTURE



Total Revenue

```
SUM (Revenue)
```

Profit Margin %

```
DIVIDE (SUM (Profit), SUM (Revenue)) * 100
```

Loyalty Ratio %

```
DIVIDE (SUM (Registered), SUM (Total)) * 100
```

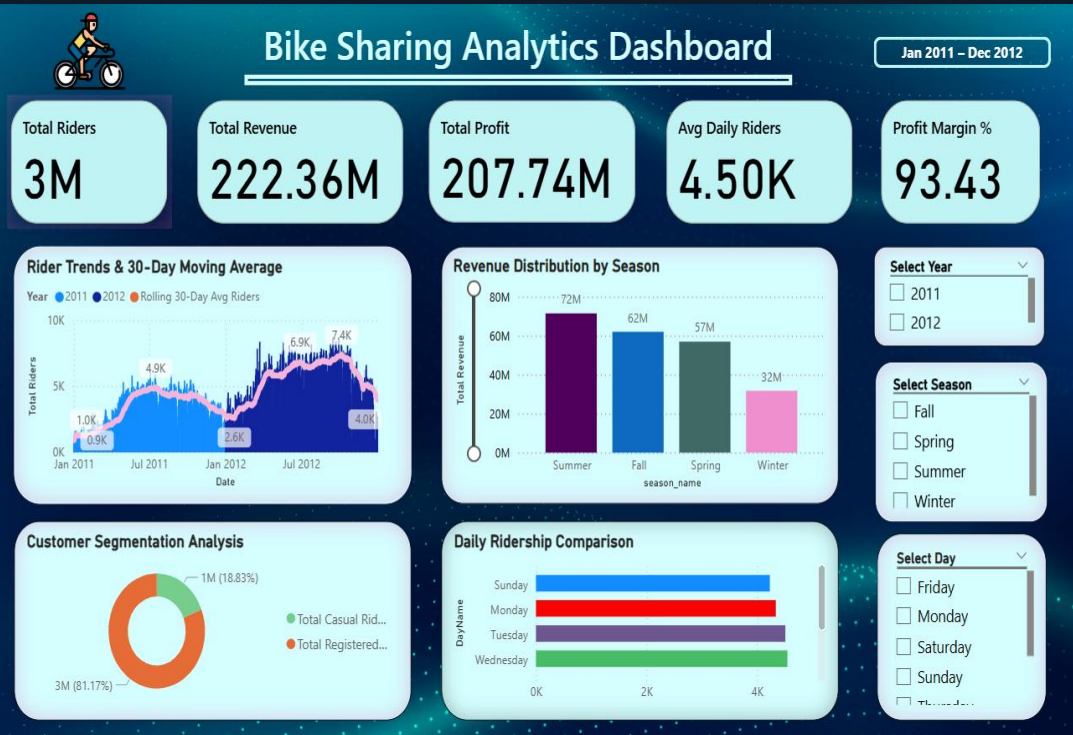
Rolling Avg (7d)

```
AVERAGEX (DATESINPERIOD (..., -7, DAY), ...)
```

YoY Growth %

```
DIVIDE ([CY Users] - [PY Users], [PY Users]) * 100
```

Page 1



Revenue KPI Cards

Total Revenue, Profit, Avg Users, Loyalty % — all dynamic with slicer context



Rider Segmentation

Donut chart comparing casual vs registered riders with loyalty ratio



Monthly Revenue Trend

Bar chart showing revenue patterns across all 12 months



Interactive Slicers

Filter by quarter, weather, day type — all visuals respond in real time

Dashboard Page 2 — Deep Dive



Weather Impact Analysis

Average users segmented by Clear, Cloudy, Rainy, Extreme — clear = highest demand



Customer Segmentation

Casual vs Registered split, loyalty trend by month, rider type classification



Revenue Distribution

Quarterly & seasonal breakdown of revenue and profit with variance analysis



Operational Insights

Day-of-week patterns, demand level distribution, high-demand day identification

What the Data Reveals



Summer = Peak Revenue **Highest**

Summer consistently generated the highest revenue across both years, driven by leisure ridership and favorable weather conditions.



Loyalty-Driven Ridership **81%+**

Registered riders contributed over 81% of total ridership — indicating a strong, loyal customer base powering consistent revenue.



Clear Weather = More Riders **62%**

Clear weather days averaged 4,876 users vs 1,803 on extreme weather days — weather is the single biggest demand driver.



Strong 2012 YoY Growth **+65%**

2012 showed approximately 65% higher ridership than 2011 — demonstrating rapid platform adoption and growing brand awareness.



Seasonal Profitability Swings **4x**

Summer profits were nearly 4x higher than winter, underscoring the need for seasonal pricing and operational adjustments.

Growth Through Complexity

⚡ CHALLENGES



Building a clean data model with proper table relationships and avoiding circular dependencies in Power BI.



Writing complex DAX measures including time intelligence functions and dynamic rolling averages.



Designing an interactive dashboard that remains intuitive while supporting multi-level filtering.



Handling encoding issues (BOM characters) in the SQL date column and ensuring correct query execution.

🎓 LEARNINGS



Mastered Power BI dashboard design including slicers, tooltips, drillthrough, and cross-filtering across pages.



Developed strong SQL skills: window functions (RANK, LAG, LEAD, rolling averages), CTEs, and view creation.



Learned business intelligence storytelling — presenting data not as numbers but as actionable decisions.



Built analytical thinking: identifying KPIs, segmenting customers, and interpreting seasonal behavior patterns.

Project Impact & Business Value

This project demonstrates a complete Business Intelligence pipeline — from raw SQL data transformation to an interactive Power BI dashboard — delivering actionable insights on rider demand, revenue trends, seasonal behavior, weather impact, and customer segmentation for a real-world bike-sharing dataset.



Full BI Pipeline

SQL → Power Query → Data Model → DAX → Interactive Dashboard



Revenue Insights

₹82.6M revenue analyzed across 730 days with profit & margin trends



Customer Analytics

Rider segmentation, loyalty analysis, and behavioral classification



Skills Showcased

SQL, Power BI, DAX, Data Modeling, Power Query, BI Storytelling



Portfolio Ready

Recruiter-facing project demonstrating real-world analytical thinking



THANK YOU



Email

vermataniya720@gmail.com



LinkedIn

linkedin.com/in/yourprofile



GitHub

github.com/yourusername

Built With:

SQL

Power BI

DAX

Power Query

Data Modeling